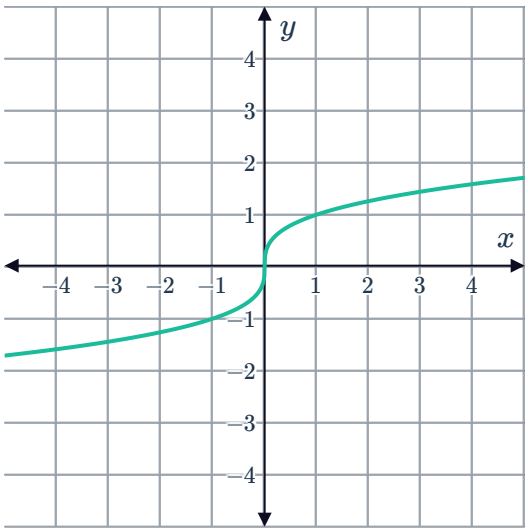
Graphs of cube root functions

1

a

x	-100	-10	-8	-3	-1	0	1	3	8	10	100
y	-4.64	-2.15	-2	-1.44	-1	0	1	1.44	2	2.15	4.64

b



c Increasing

d No, x can be any real number.

e All real y.

f No

g No

h ∞

i As x increases, the function increases at a slower and slower rate.

j As x increases, the function increases at a faster and faster rate.

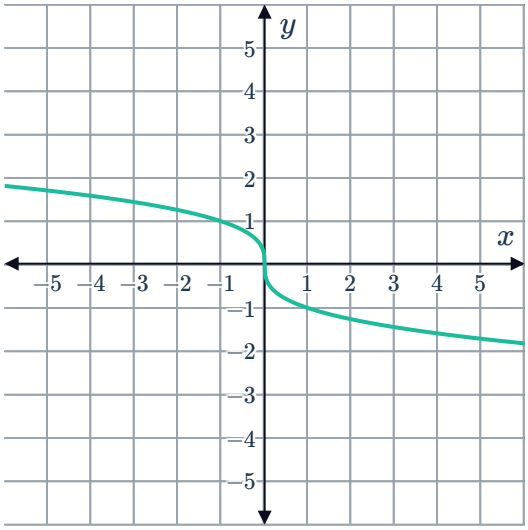
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2

a

x	-100	-10	-8	-3	-1	0	1	3	8	10	100
y	4.64	2.15	2	1.44	1	0	-1	-1.44	-2	-2.15	-4.64

b



c Yes

d Yes

e All real All real x.

f all real y

g Decreasing function

h As x increases, the function decreases more and more rapidly up to $x = 0$, and from $x = 0$ onwards, the rate of decrease slows down.

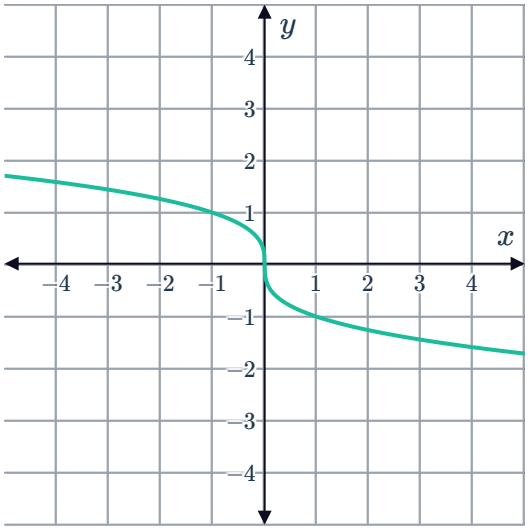
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3

a

x	-100	-10	-8	-3	-1	0	1	3	8	10	100
y	4.64	-2.15	-4.64	2.15	2	1.44	1	0	-1	-1.44	-2

b



c Decreasing

d All real x

e All real y

f ∞

g $-\infty$

4

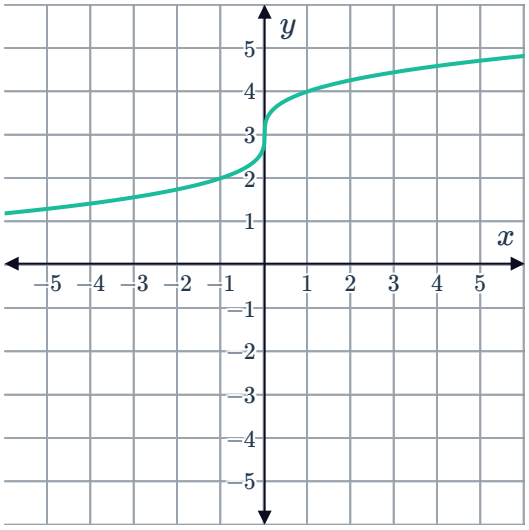
a Yes

b ∞

c $(0, 3)$

d 1

e



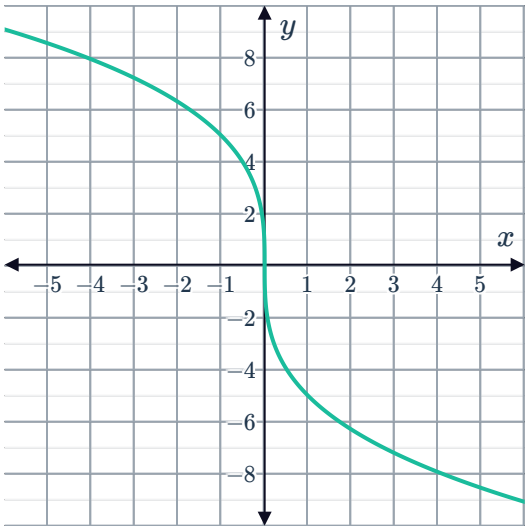
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5

a All real x.

b All real y.

c

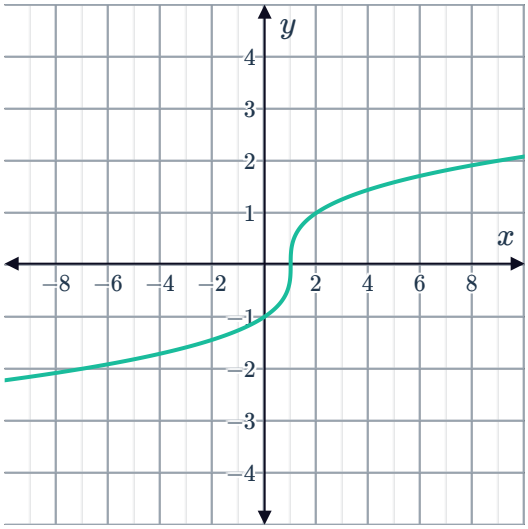


6

a

x	-7	0	1	2	9
$\sqrt[3]{x-1}$	-2	-1	0	1	2

b



c All real numbers.

d All real numbers.

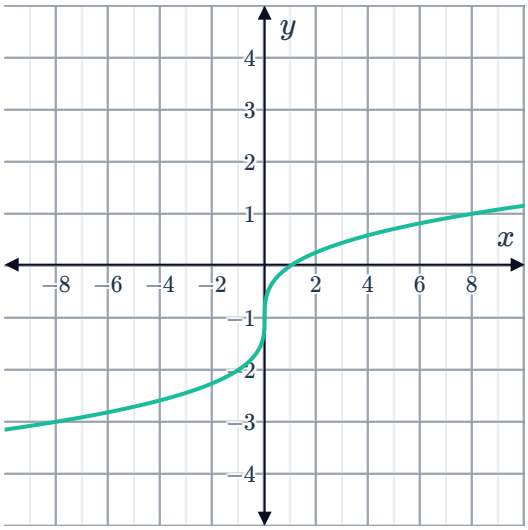
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7

a

x	-8	-1	0	1	8
$\sqrt[3]{x}-1$	-3	-2	-1	0	1

b



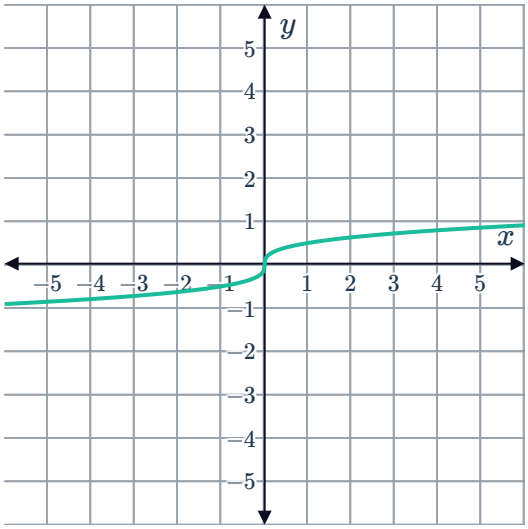
c All real numbers.

d All real numbers.

8

a

i

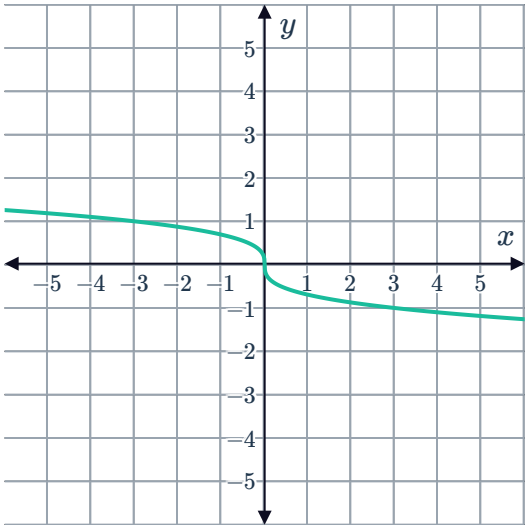


ii All real numbers.

iii All real numbers.

b

i



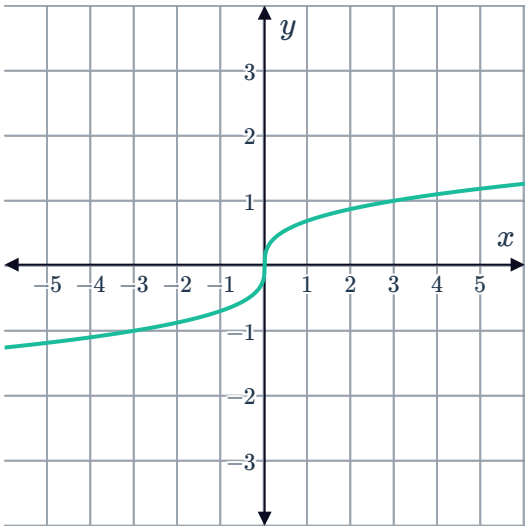
ii All real numbers.

iii All real numbers.

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c

i

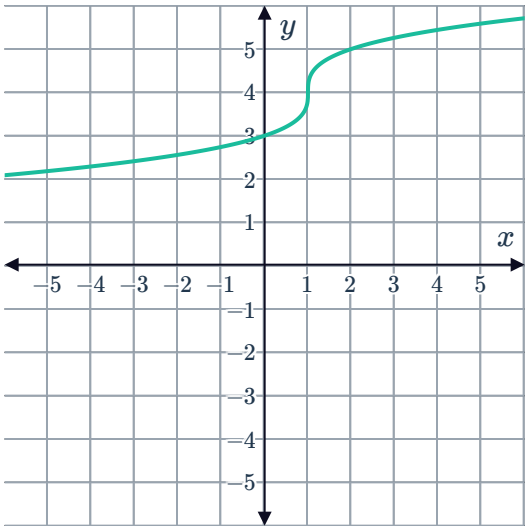


ii All real numbers.

iii All real numbers.

d

i

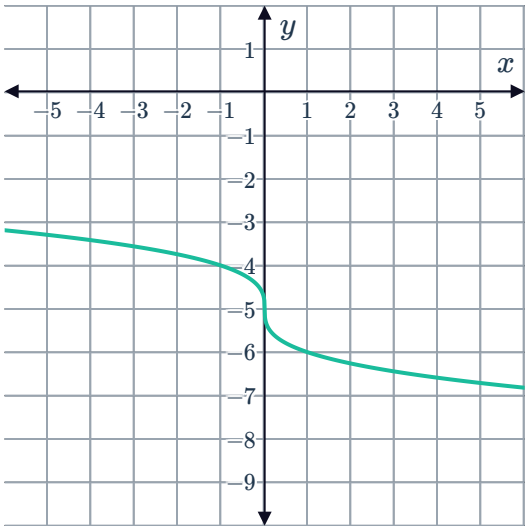


ii All real numbers.

iii All real numbers.

e

i

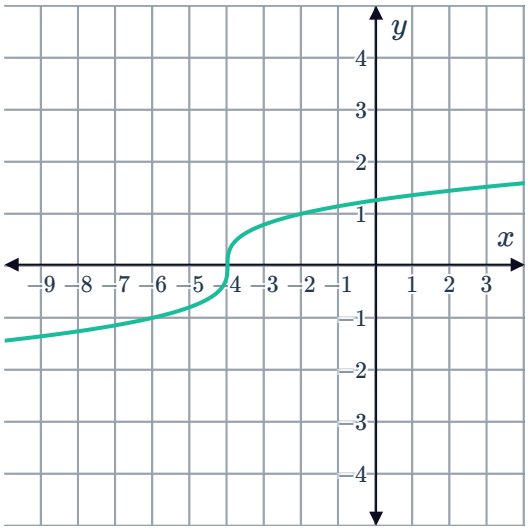


ii All real numbers.

iii All real numbers.

f

i



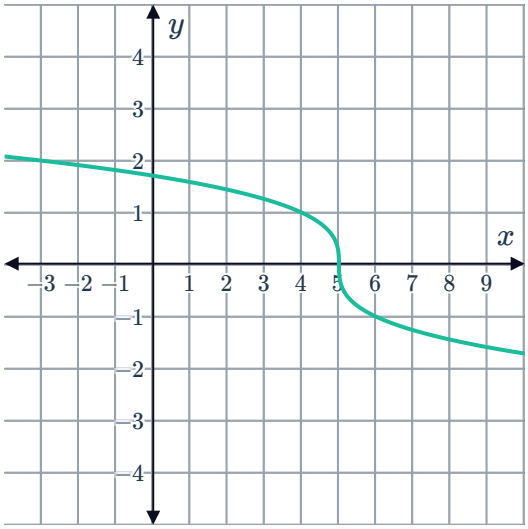
ii All real numbers.

iii All real numbers.

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g

i

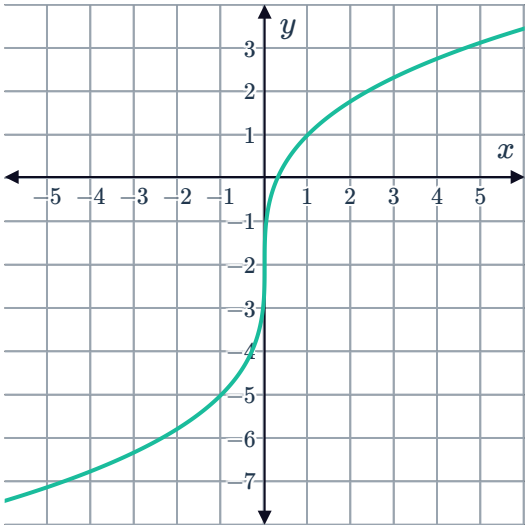


ii All real numbers.

iii All real numbers.

h

i

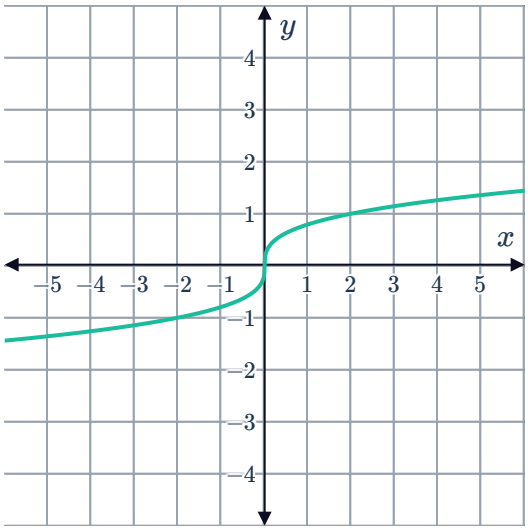


ii All real numbers.

iii All real numbers.

i

i

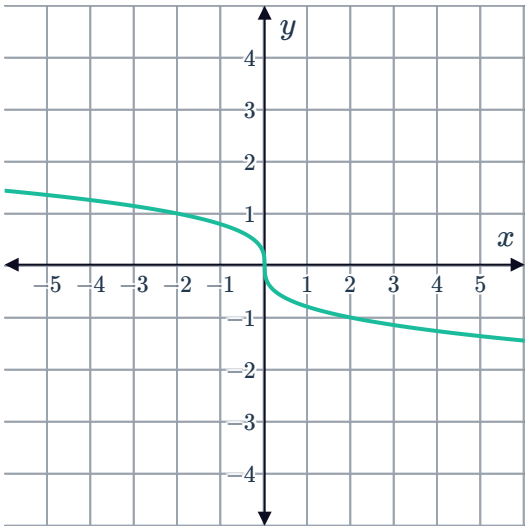


ii All real numbers.

iii All real numbers.

j

i



ii All real numbers.

iii All real numbers.

Characteristics of cube root functions

9 $y = 5\sqrt[3]{x}$ increases more rapidly than $y = \sqrt[3]{x}$.

10 All real x and all real y.

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11

a All real x.

b All real y.

c The functions $y = \sqrt[3]{x}$ and $y = \sqrt[3]{x - 4}$ increase at the same rate.

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